

▼ Titanic Data Visualization Dashboard

Maincrafts Technology Internship – Task 4

Objective

Build a mini visualization dashboard using Python to communicate important insights from the Titanic dataset.

Dashboard Includes

- Data Cleaning
- Feature Engineering
- Histogram
- Bar Chart
- Box Plot
- Scatter Plot
- Correlation Heatmap
- Final Insights

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set_style("whitegrid")
```

▼ Load Dataset

```
df = pd.read_csv("train (1).csv")
```

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S

Insight

The Titanic dataset contains passenger details such as age, sex, passenger class, fare, embarkation port, and survival status. These variables will be used to create meaningful visualizations.

▼ Dataset Information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
# 0  PassengerId      891 non-null    int64
# 1  Survived         891 non-null    bool
# 2  Pclass           891 non-null    int64
# 3  Name             891 non-null    object
# 4  Sex              891 non-null    object
# 5  Age              891 non-null    float64
# 6  SibSp            891 non-null    int64
# 7  Parch            891 non-null    int64
# 8  Ticket           891 non-null    object
# 9  Fare             891 non-null    float64
# 10 Cabin          294 non-null    object
# 11 Embarked       891 non-null    object
```

```
0 PassengerId 891 non-null int64
1 Survived    891 non-null int64
2 Pclass      891 non-null int64
3 Name        891 non-null object
4 Sex         891 non-null object
5 Age         714 non-null float64
6 SibSp       891 non-null int64
7 Parch       891 non-null int64
8 Ticket      891 non-null object
9 Fare        891 non-null float64
10 Cabin      204 non-null object
11 Embarked   889 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
print("Dataset Shape:", df.shape)
```

```
Dataset Shape: (891, 12)
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

```
dtype: int64
```

Insight

The dataset contains missing values in the Age, Cabin, and Embarked columns. These missing values need to be handled before visualization.

▼ Data Cleaning

```
df['Age'] = df['Age'].fillna(df['Age'].median())
```

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
df = df.drop(columns=['Cabin'], errors='ignore')
```

```
df.isnull().sum()
```

```
0
PassengerId 0
Survived 0
Pclass 0
Name 0
Sex 0
Age 0
SibSp 0
Parch 0
Ticket 0
Fare 0
Embarked 0

dtype: int64
```

Insight

Missing values have been successfully handled. The Age column was filled using the median, Embarked using the mode, and the Cabin column was removed because it contained a large number of missing values.

Feature Engineering

```
df['FamilySize'] = df['SibSp'] + df['Parch']
```

```
df['AgeGroup'] = pd.cut(df['Age'], bins=[0,12,18,30,50,80], labels=['Child','Teen','Young Adult','Adult','Senior'])
```

```
df[['Age','AgeGroup','FamilySize']].head()
```

	Age	AgeGroup	FamilySize
0	22.0	Young Adult	1
1	38.0	Adult	1
2	26.0	Young Adult	0
3	35.0	Adult	1
4	35.0	Adult	0

Insight

Two new features were created to improve the analysis:

- FamilySize combines siblings/spouses and parents/children.
- AgeGroup categorizes passengers into meaningful age ranges for easier comparison.

Dashboard Visualizations

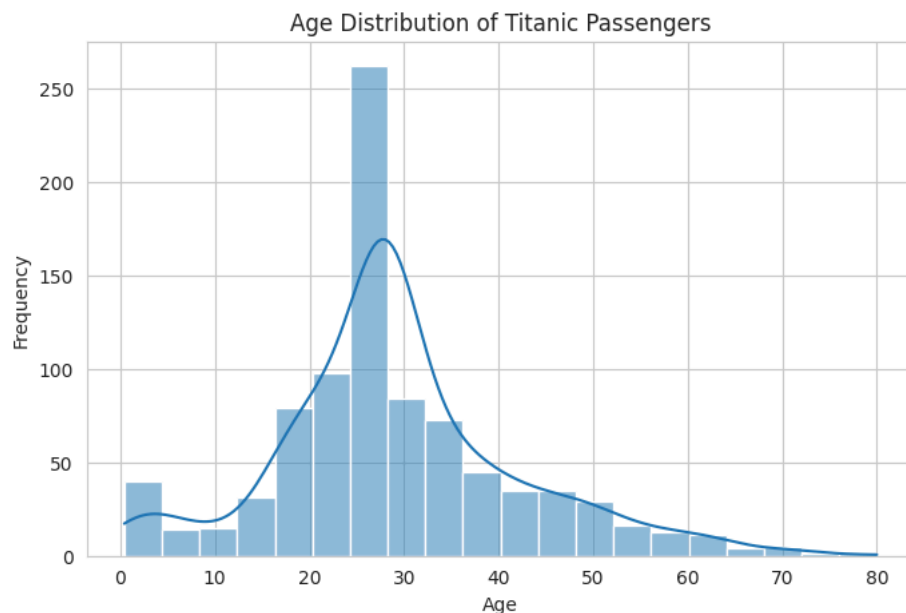
1. Age Distribution of Passengers

```
plt.figure(figsize=(8,5))

sns.histplot(df['Age'], bins=20, kde=True)

plt.title("Age Distribution of Titanic Passengers")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```



Insight

The histogram shows the distribution of passenger ages. Most passengers were young adults, while fewer passengers belonged to the elderly age group.

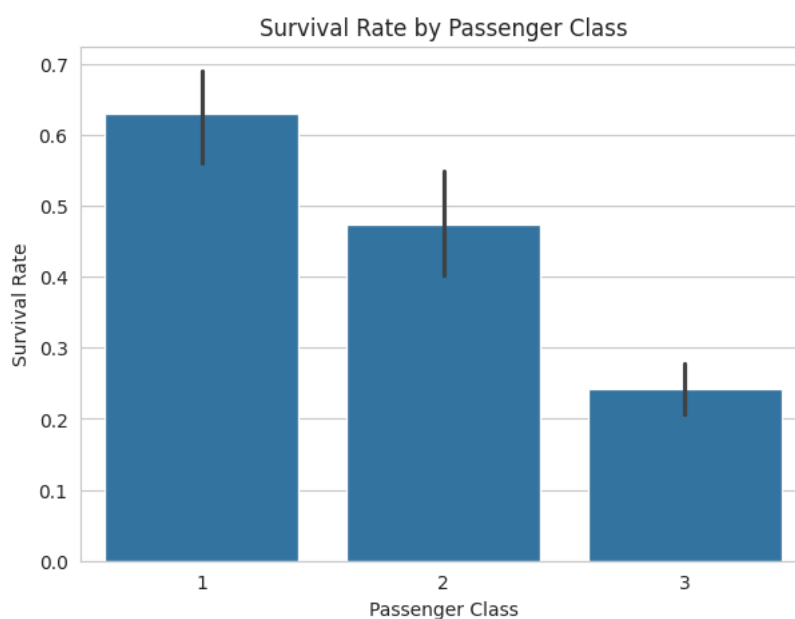
2. Survival Rate by Passenger Class

```
plt.figure(figsize=(7,5))

sns.barplot(x='Pclass', y='Survived', data=df)

plt.title("Survival Rate by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Survival Rate")

plt.show()
```



Insight

Passengers travelling in First Class had a higher survival rate compared to those in Second and Third Class.

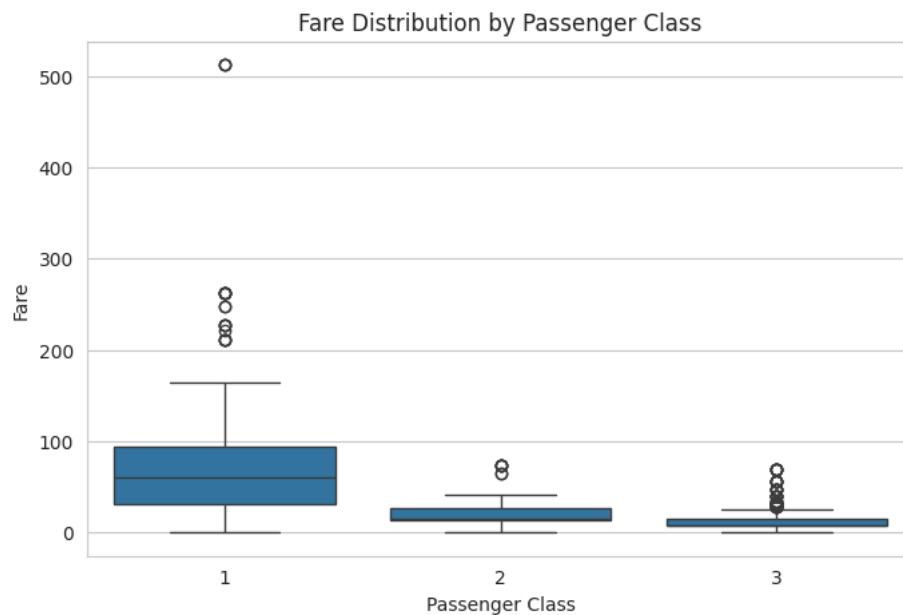
3. Fare Distribution by Passenger Class

```
plt.figure(figsize=(8,5))

sns.boxplot(x='Pclass', y='Fare', data=df)

plt.title("Fare Distribution by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Fare")

plt.show()
```

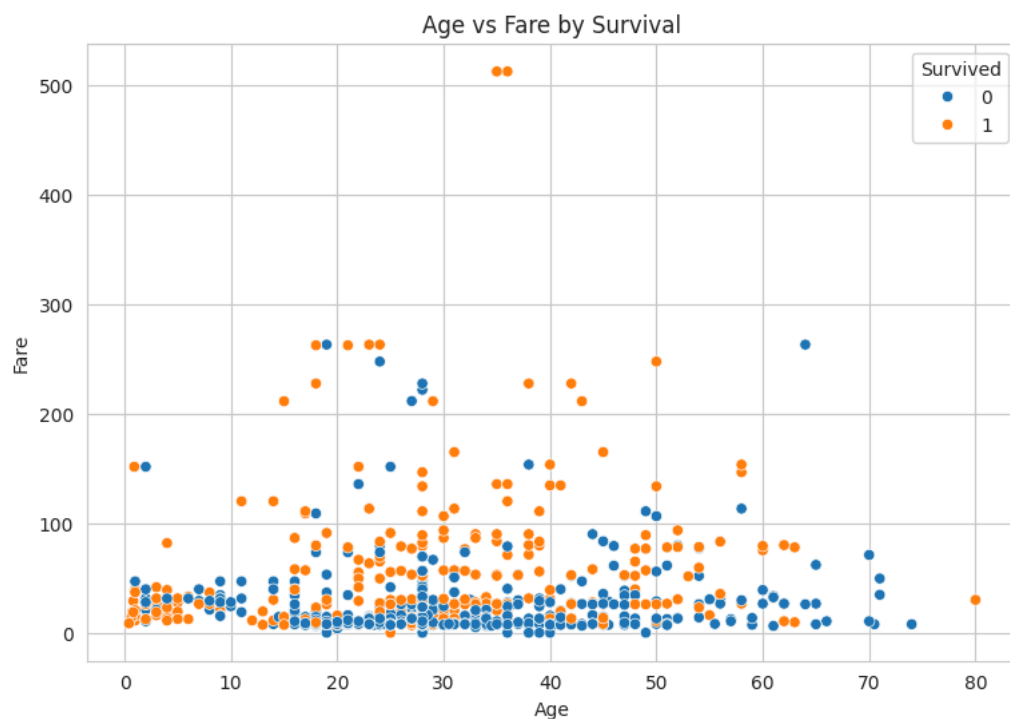


Insight

The boxplot shows that First Class passengers generally paid higher fares. It also highlights the presence of outliers in ticket prices.

4. Age vs Fare (Colored by Survival)

```
plt.figure(figsize=(9,6))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```



Insight

The scatter plot illustrates the relationship between passenger age and fare, with colors indicating survival status. It helps identify patterns among survivors and non-survivors.

5. Correlation Heatmap

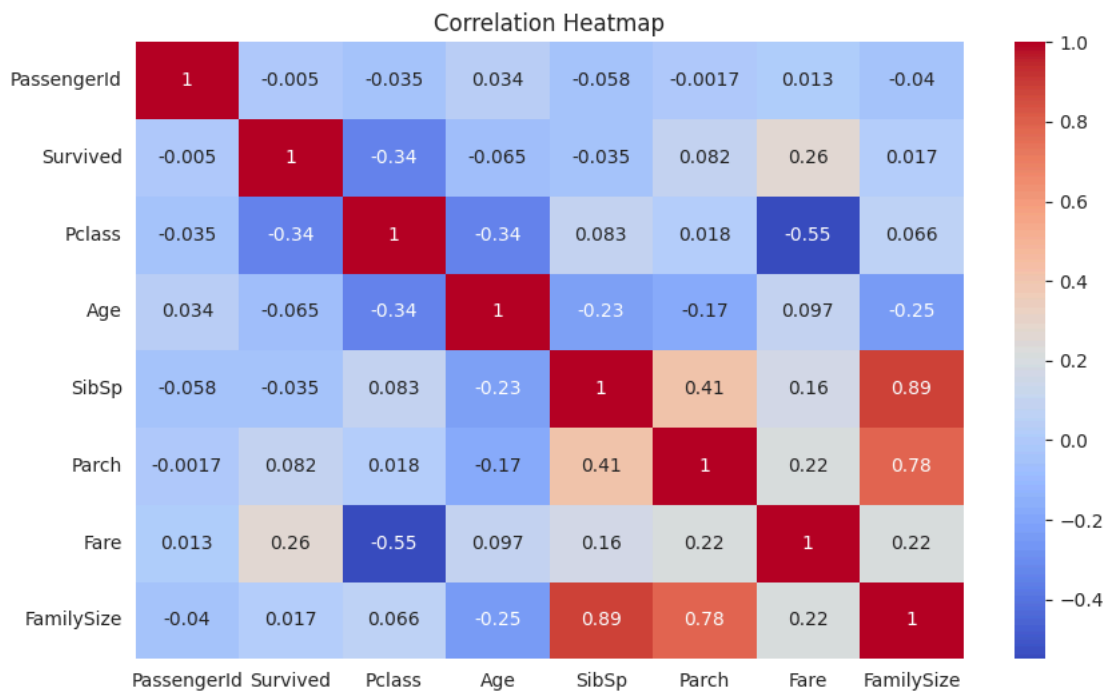
```
plt.figure(figsize=(10,6))

corr_matrix = df.select_dtypes(include='number').corr()

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()
```



Insight

The heatmap displays the correlation between numerical features. Strong positive or negative correlations help identify relationships that influence passenger survival.

Dashboard Summary

Key Findings

- Most passengers were young adults.
- First Class passengers had the highest survival rate.
- Higher ticket fares were generally associated with First Class.
- Age and Fare showed varying survival patterns.
- The heatmap revealed important relationships among numerical variables.

Conclusion

This dashboard successfully communicates key insights from the Titanic dataset through multiple visualizations.

The project involved:

- Data Cleaning

- Feature Engineering
- Exploratory Data Analysis (EDA)
- Dashboard Creation
- Data Storytelling

The visualizations helped explain survival patterns across passenger class, age, fare, and family size, demonstrating how charts can transform raw data into meaningful business insights.